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Seasonal and interannual variations of the oceanic heat flux under a landfast Antarctic sea ice cover

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Abstract. A multilayer thermodynamic model is used to simulate sea ice growth for 12 years between 1958 and 1986 in the vicinity of the Australian station Mawson on the coast of East Antarctica. The atmospheric forcing data for the model are derived from radiosonde profiles and from surface measurements. Global radiation data are available for 4 years, and we use these measurements for comparison with the results of a Zillman-type model for global radiation. Combining the thermodynamic model with sea ice thickness measurements for 12 years, we solve the energy balance equation for the oceanic heat flux. The oceanic heat flux is not constant but changes with time within the year and from year to year. The oceanic heat flux averages 7.9 W/m^2, and the yearly means vary between 5 and 12 W/m^2. Seasonal values of the oceanic heat flux range from 0 to 18 W/m^2. From the yearly averaged values a decadal trend is evident: During the first years that were analyzed the yearly average lies well above 10 W/m^2; then in the mid-1970s a decrease to 9 W/m^2 occurs, while for all later years the values are ~6–8 W/m^2. In general, the oceanic heat flux increases from the start of the fast ice formation season in early April until it breaks out in December or January. To compare the calculated oceanic heat fluxes for different years, we divide the total ice season into three characteristic time regimes of the sea ice growth and calculate the averaged oceanic heat fluxes for each regime. For the first regime (through August) the mean flux is 2.7 W/m^2, for the middle regime (September) it is 8.4 W/m^2, and for the final regime (October–January) it is 17 W/m^2. We discuss the results of our model calculations in conjunction with current meter observations, which give evidence of seasonally varying intrusions of relatively warm Circumpolar Deep Water into Prydz Bay. Comparison of passive microwave data of sea ice extent and concentration (from the scanning multichannel microwave radiometer sensor) with the model results reveals a correlation between the magnitude of the oceanic heat flux and local features such as polynyas.

1. Introduction

The oceanic heat flux exerts a large influence on the formation and the thickness of sea ice. Global climate models have proved to be highly sensitive to the oceanic heat flux. While for the Arctic an oceanic heat flux of 2 W/m^2 is widely considered to be a reasonable annual average value [*Semtner*, 1976; *Maykut and Perovich*, 1987], much higher values are generally used for the Antarctic. *Parkinson and Washington* [1979] found best results for their large-scale model by using a value of 25 W/m^2; this value was applied as a constant throughout the year. *Gordon and Huber* [1990] derive a mean oceanic heat flux of 16 W/m^2 for the section of the Greenwich meridian between 60° and 70°S. *Gordon* [1981] contended that a heat flux as high as 30 W/m^2 was needed to explain the rapid spring dissipation of the ice cover. The related upwelling is most likely when a weak pycnocline exists as a result of low freshwater input. Hence the seasonality of southern sea ice may largely be determined by the magnitude of freshwater input from melting ice, precipitation, and glaciers. *Parkinson and Good* [1982] showed a strong dependence of the modeled ice extent and thickness in the Weddell Sea on heat flux. In contrast to these high oceanic heat fluxes, *Hibler and Ackley* [1982], using a thermodynamic/dynamic sea ice model [*Hibler*, 1979], were able to match the Weddell Sea annual ice cycle with a heat flux as low as 2 W/m^2 but with ice advection playing an important role. Studies by *Robertson et al.* [1995] and *Lytle and Ackley* [1996] have shown that the western Weddell Sea has an oceanic heat flux of 2–7 W/m^2, which may be one of the reasons why this region has a perennial sea ice cover and why the ice thickness relative to other regions is higher.

The importance of a seasonally varying oceanic heat flux in thermodynamic models for the growth of sea ice has been handled varyingly in previous studies. While *Parkinson and Washington* [1979] saw the influence of a seasonally varying

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oceanic heat flux on the growth of the sea ice, *Harvey* [1988] found that the seasonal variation does not influence the minimum ice thickness but that the maximum ice thickness and the sea ice area are significantly affected by seasonal changes in the oceanic heat flux. *Allison* [1981] studied the fast ice cover around the Antarctic station Mawson and found two peaks in the oceanic heat flux during the growth season. He proposed that for the initial ice growth the oceanic heat flux shows a maximum, which is caused by the ice growth itself, since a rapid growth of sea ice sets up a thermohaline convection, which deepens the mixed layer by pumping warmer water upward. The second lesser peak, which occurred around the time of maximum ice thickness, was attributed to larger-scale oceanic processes.

In this study, the oceanic heat flux underneath the seasonal fast ice cover near Mawson and Davis is calculated using a thermodynamic ice model. The model, discussed in section 2, is forced by the surface energy fluxes, which are derived from measured atmospheric parameters, such as air temperature, wind speed, and relative humidity. These surface fluxes are used in conjunction with the measured fast ice thickness to solve for the oceanic heat flux. The atmospheric data set and the ice thickness measurements on the fast ice near Mawson are described in section 3. The model's sensitivity to different parameters is outlined in section 4. We then calculate the oceanic heat flux from the thermodynamic model in combination with the ice thickness measurements.

2. Sea Ice Model

The one-dimensional thermodynamic model for the growth and decay of sea ice is adopted from *Semtner* [1976]. The driving atmospheric fluxes are calculated from actual 3-hourly or 12-hourly measurements. For each time step, vertical profiles of temperature, salinity, and thermal conductivity within the ice are calculated.

A one-layer approach is used until an ice thickness z_{ice} of 0.1 m is exceeded. As the ice thickens, the whole ice column is divided into a maximum of 12 layers of nearly equal thickness. In cases where snow overlies the ice the snow is considered as a single layer. At the snow-ice interface and at the ice-ice intrafaces, continuity equations for the thermal conductivity and for the conductive heat flux apply.

2.1. Thermodynamic Model

In the absence of surface melt the heat balance at the ice-atmosphere interface can be written as

$$(1-\alpha)F_{SWin} - I_0 + F_{LWin} - F_{LWout} + F_{SH} + F_{LH} = F_c. \quad (1)$$

Here F_{SWin} is the incoming short wave radiation, α is the surface albedo, I_0 is the net flux of radiative energy, which is absorbed in the interior of the sea ice, F_{LWin} is the incoming long wave radiation, F_{LWout} is the outgoing long wave radiation, F_{SH} is the sensible heat flux, F_{LH} is the latent heat flux, and F_c is the conductive heat flux. The net flux at the upper surface is balanced by a conductive heat flux within the sea ice. The surface temperature T_{sur} is iteratively determined from (1).

On the lower surface of the ice another heat balance equation has to be satisfied by the conductive flux:

$$F_{ocean} + \rho_{ice} L_f \frac{\delta z_{ice}}{\delta t} = F_{c^*}. \quad (2)$$

Here L_f is the latent heat of fusion of the sea ice, ρ_{ice} is the bulk density of the sea ice, $\delta z_{ice}/\delta t$ is for the growth rate, F_{ocean} is the vertical heat flux from the ocean, and F_{c^*} is the conductive heat flux at the bottom of the ice.

2.1.1. Short wave radiation. The incoming global radiation is calculated with a Zillman-type model [*Zillman*, 1972], which is modified [*Shine*, 1984] for high-latitude effects such as enhanced scattering. The incoming global radiation for cloud-free conditions F_{SW0} is dependent on latitude, day, and time:

$$F_{SW0} = \frac{S_c \cos^2 A}{(\cos A + 1.0)e_a \times 10^{-5} + \cos A + 0.046}. \quad (3)$$

Here S_c is the solar constant, A is the solar altitude, a function of latitude, day, and local time, and e_a is the partial water vapor pressure of the atmosphere in pascal. A correction for the cloudiness is applied by parameterizing the loss through scattering [*Laevastu*, 1960]:

$$F_{SWin} = F_{SW0}(1.0 - 0.6c^3), \quad (4)$$

where c is the observed total cloudiness (in tenths).

To obtain daily average data, we integrate the global radiation over a full 24 hours. Since the albedo of different surface types varies significantly, we use the following values [*Weller*, 1972; *Perovich et al.*, 1986]: (1) open water and leads ($\alpha = 0.10$), (2) ice with snow cover ($\alpha = 0.82$), (3) thick ice ($z_{ice} \geq 1.0$ m) without snow cover ($\alpha = 0.73$), (4) thin ice ($z_{ice} < 1.0$ m) without snow cover ($\alpha = 0.338 + 0.602 z_{ice} - 0.488 z_{ice}^2 + 0.127 z_{ice}^3$), and (5) melting ice or snow ($\alpha = 0.6$). For snow-free ice a certain percentage of the incoming short wave radiation I_0 penetrates into the ice and leads to heat storage in the top layers of the ice column [*Maykut*, 1986; *Ebert and Curry*, 1993]. For the Arctic the chosen percentage of penetrating short wave radiation lies between 18 and 35% [*Grenfell and Maykut*, 1977]. The ice conditions between the Arctic and Antarctic differ considerably; for example, the ice thickness is generally less, melt ponds are rarely found, and the backscatter coefficient is higher in the Antarctic. To account for Antarctic conditions, we choose a value of 17% of the incoming short wave radiation to penetrate into the ice:

$$I_0 = 0.17(1-\alpha)F_{SWin}. \quad (5)$$

The amount of short wave radiation penetrating into the ice interior is stored, and as a result, the ice temperature rises.

2.1.2. Long wave radiation. The outgoing long wave radiation F_{LWout} is calculated using the Stefan-Boltzman law for a gray body with an effective emissivity ε_{sur} (0.97 for ice, leads, and melt ponds or 0.99 for snow):

$$F_{LWout} = \varepsilon_{sur}\sigma T_{sur}^4. \quad (6)$$

Here σ is the Stefan-Boltzman constant, and T_{sur} is the surface temperature of the ocean, ice, or snow.

The incoming long wave radiation F_{LWin} is modified by the scattering and reflection on clouds and particles in the atmosphere. Similar to the incoming short wave radiation, F_{LWin} is decreased by the cloud amount c:

$$F_{LWin} = (1 - nc)\varepsilon_0 \sigma T_{air}^4. \quad (7)$$

Here n is given by an empirical constant, equal to 0.275 [*Schwerdtfeger*, 1984], and T_{air} is the air temperature. The emissivity of the air ε_0 can be calculated as a function of the air temperature [*Idso and Jackson*, 1969]:

$$\varepsilon_0 = 1 - 0.261e^{-7.77\times10^{-4}(T_{air}-273)^2}. \quad (8)$$

2.1.3. Turbulent heat fluxes. Differences in temperature and water vapor content between atmosphere and ice surface result in turbulent transport of sensible and latent heat in the lower atmosphere. Both are smaller than the radiation fluxes, but they are of the same magnitude as the net radiation and are significant for the heat balance.

The sensible heat flux is a function of the air density ρ_{air}, the specific heat at constant pressure c_p, the transfer coefficient of sensible heat ($C_s = 1.7 \times 10^{-3}$), the wind speed u, and the difference between air temperature T_{air} and surface temperature T_{sur} [*Maykut*, 1986]:

$$F_{SH} = \rho_{air} c_p C_s u (T_{air} - T_{sur}). \quad (9)$$

The flux of latent heat depends on the transfer coefficient of latent heat ($C_e = 1.65 \times 10^{-3}$), the latent heat of vaporization or freezing L_f, and the difference in the specific humidity between a height of 10 m q_{air} and directly above the ice surface q_{sur}:

$$F_{LH} = \rho_{air} C_e L_f u (q_{air} - q_{sur}). \quad (10)$$

The latent heat of sea ice is a function of the ice temperature T_{ice} and the ice salinity S_{ice} [*Yen*, 1981]:

$$L_f = 4.19 \times 10^3 \left[79.68 - 0.505(T_{ice} - 273.15) - 27.3 S_{ice} + 4311.5 \frac{S_{ice}}{T_{ice} - 273.15} \right]. \quad (11)$$

2.1.4. Precipitation. The model does not calculate the thickness of accumulated snow on the ice. There are no measurements of precipitation available, but the precipitation rate is rather low, and snow is often quickly blown away. Snow thickness was usually measured with the ice thickness, and we use the measured snow thicknesses directly in the model.

2.2. Growth and Decay of Sea Ice

The model is initialized on the first day of recorded ice formation. The thickness of the sea ice and the vertical temperature profile within the sea ice are governed by the energy balances on both the upper and lower surfaces and by the conduction of heat through the ice column. The heat conduction equation can be written as

$$\rho_{ice} c_{ice} \frac{\delta T}{\delta t} = \kappa_{ice} \frac{\delta^2 T}{\delta z^2}. \quad (12)$$

Here the product of the bulk ice density ρ_{ice} and the specific heat capacity c_{ice} of the ice is the volumetric heat capacity. The value κ_{ice} is the thermal conductivity of the ice layer with a thickness z [*Yen*, 1981]. A similar equation is used where there is a snow cover. The conductive heat flux F_c is

$$F_c = \kappa_{ice} \frac{\delta T}{\delta z}. \quad (13)$$

This conductive heat flux F_c transports heat from the relatively warm bottom to the colder surface of the ice.

The above equations are applied separately to each of the layers using the individual layer values of the thermal conductivity. The thermal conductivity of each layer is parameterized as a function of the layer temperature and the brine volume and hence the layer salinity [*Cox and Weeks*, 1983]. Temperature and salinity profiles from the previous time step are used for this parameterization.

2.3. Salinity Model

In the model, different processes of brine transport (gravity drainage and brine expulsion) are calculated individually for each layer. We use the equations of *Cox and Weeks* [1988] to describe the changes in salinity with time. Brine drainage transports salt from the top layers of the sea ice into the bottom layers and from there into the ocean. The loss of salinity per time step dS_{ice}/dt is a function of the vertical temperature gradient dT_{ice}/dz and the brine volume ν_{bri}:

$$\frac{dS_{ice}}{dt} = 1.68 \times 10^{-5} \frac{dT_{ice}}{dz} - 3.37 \times 10^{-7} \nu_{bri} \frac{dT_{ice}}{dz}. \quad (14)$$

The temperature decrease of the ice during growth and the associated freezing of ice in the cells cause brine expulsion:

$$\frac{S_{ice}(T_2)}{S_{ice}(T_1)} = \left[\frac{S_{bri}(T_2)}{S_{bri}(T_1)} \right]^{(1-1/\rho_{ice})} \frac{\rho_{bri}(T_2)}{\rho_{bri}(T_1)} e^{c/\rho_{ice}(S_{bri}(T_1) - S_{bri}(T_2))}. \quad (15)$$

Here S_{ice} is the salinity of the sea ice at a given temperature, S_{bri} is the salinity of the entrapped brine at a certain temperature, ρ_{ice} is the density of pure ice (918 kg/m^3), ρ_{bri} is the brine density, and c is a constant describing the brine density gradient with temperature (0.8 kg/(m^3 K)). At each time step the two equations of brine transport are combined with the salinity history of each layer to get the salinity profile of the sea ice slab.

2.4. Oceanic Heat Flux

Within the numerical model the oceanic heat flux is calculated at each time step. We iteratively solve (1) for the surface temperature; we then use (12) and (13) to solve for the conductive heat flux in each layer. This conductive heat flux along with the measured ice growth is used to solve (2). Values of the oceanic heat flux are obtained as the residual of the energy balance combined with the ice thickness measurements.

3. Input Data

3.1. Meteorological Observations

One aim of this study was to force the thermodynamic model with as much actual data as possible. From the meteorological observations at the weather stations at Mawson and Davis, 3-hourly surface and 12-hourly radiosonde data are available. From the radiosonde we use the air temperature and the relative humidity measurements at 2- and 50-m levels to interpolate data to 10 m; the surface measurements provide us with surface air pressure, wind speed, and direction at 10-m elevation, cloudiness, and visibility. This meteorological data set and derived parameters enable us to calculate all the necessary radiation and turbulent fluxes at the surface.

The measurement site for the meteorological parameters is uphill from the site where the sea ice measurements were made. Hence the air temperature and wind speed may be slightly different than those at the actual sea ice site. The effect of this is examined as part of a sensitivity study. The global radiation was measured comprehensively in 1965. Further measurements are available for 1969, 1976, and 1977.

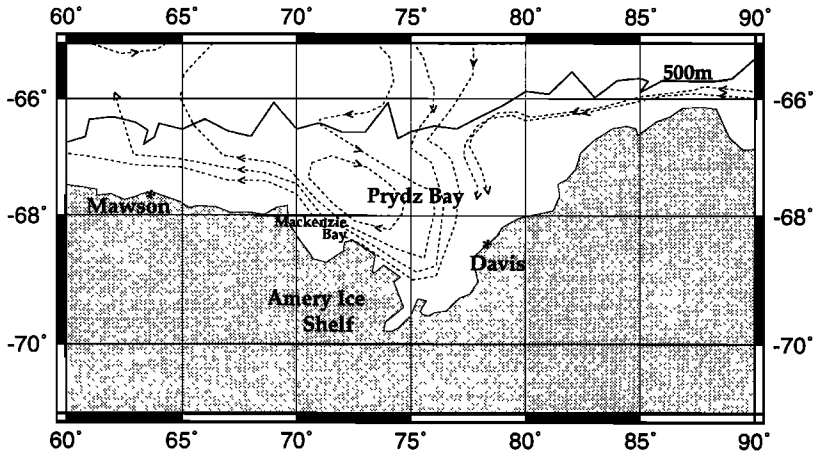

Figure 1. Coastline of East Antarctica in the Prydz Bay region. The 500-m bathymetric line and geostrophic contour lines (dashed), adapted from geopotential anomaly contours of *Nunes Vaz and Lennon* [1996], are shown.

3.2. Fast Ice Thickness at Mawson and Davis

Measurements of sea ice thickness at Australian coastal Antarctic stations have been made intermittently since the first Australian National Antarctic Research Expeditions (ANARE) station was established in Antarctica in 1954. Thickness measurements were made in freshly drilled 5-cm-diameter holes in the ice, using a tape with a cross bar at the lower end. A separate hole was usually drilled for each observation but always within a few meters of the observing site which was usually marked with a cane. Accuracy of the measurements is estimated at ±2 cm. The observers also recorded dates of ice formation, breakout, and other major phenomena. In addition, the thickness of the overlying snow was measured. The measurements that were made during 1969 at Mawson were the most extensive ones.

We use observations collected at Mawson between 1958 and 1986. All observations are of undeformed landfast sea ice, which remains attached to the coast or islands throughout the season. Measurements from Mawson are from a site a few hundred meters offshore in West Bay. Water depth at this site is over 300 m, and the bay is connected via a deep glacial canyon, Kista Strait, to the continental shelf north of coastal islands. Kista Strait runs north northeast and has a water depth >300 m; this allows a penetration of most of the shelf water column to the coastline at Mawson. The 500-m bathymetric contour on Figure 1 shows the deep indention of the contour toward Mawson.

Frazil and pancake ice start to form in Mawson waters from late February to early March, and a continuous ice cover has usually formed by late March. The climate of this area is characterized by persistent and strong offshore katabatic winds which may delay ice formation in "wind pools" (typically of tens to hundreds of meters in dimension) along the coast. The winds usually keep the sea ice close to the shoreline clear of snow, while offshore islands help to hold the sea ice in place. Initial thin ice covers may be dissipated by storm events, but with few exceptions a stable fast ice cover exists from mid-April onward. Stable landfast ice usually persists throughout the winter for a hundred kilometers both east and west of Mawson. Due north of Mawson the fast ice extends for tens of kilometers. There is no platelet ice found at the measurement sites around Mawson.

Maximum ice thickness, which can vary from 1.3 to nearly 2.0 m, is attained around early October. The final breakout of the seasonal fast ice near Mawson usually occurs in mid to late January, during high tide and strong winds. Breakout is often induced by resupply ships breaking the last few kilometers of ice into the harbor in January. Figure 2 illustrates the distribution of ice formation and growth at Mawson. Table 1 lists the occurrence of the breakout events and the growth statistics for the ice cover, which forms after initial breakouts.

A limited number of sea ice observations are also available from Davis. At Davis, ice thickness measurements were made ~1 km north of the station in water of <10 m depth. Water depths immediately around Davis and the offshore islands are much shallower than around Mawson. Therefore we expect little warm water originating from deep water to surface around Davis, a low oceanic heat flux, and little impact on the ice growth. Davis is on the edge of the Vestfold Hills and does not experience katabatic winds; consequently, a thick snow cover usually develops on the ice. Ice starts to form at Davis from early March, and a stable cover usually persists within the immediate vicinity of the station from April onward. For the 6 years of data from this site the maximum ice thickness was 1.6–1.8 m, usually attained in early November. Final breakup of the fast ice usually occurs during high tide and strong winds from late December to January.

4. Model Sensitivity

We estimate the model's sensitivity to the forcing parameters by calculating changes to the maximum ice thickness for a given change in the investigated parameter. At any time only one parameter is investigated while the others are held at their standard values.

Global radiation data are only available for the years 1965, 1969, 1976, and 1977. The quality of radiation measurements depends very much on the care given to the instrumental environment. From notes accompanying the Mawson measurements we conclude that large error bars (±30%) apply in most years, but we use these measurements to test the performance of the Zillman-type radiation model (Figure 3). The summer maximum of the modeled global radiation is less than that measured, but in spring and autumn the model gives higher

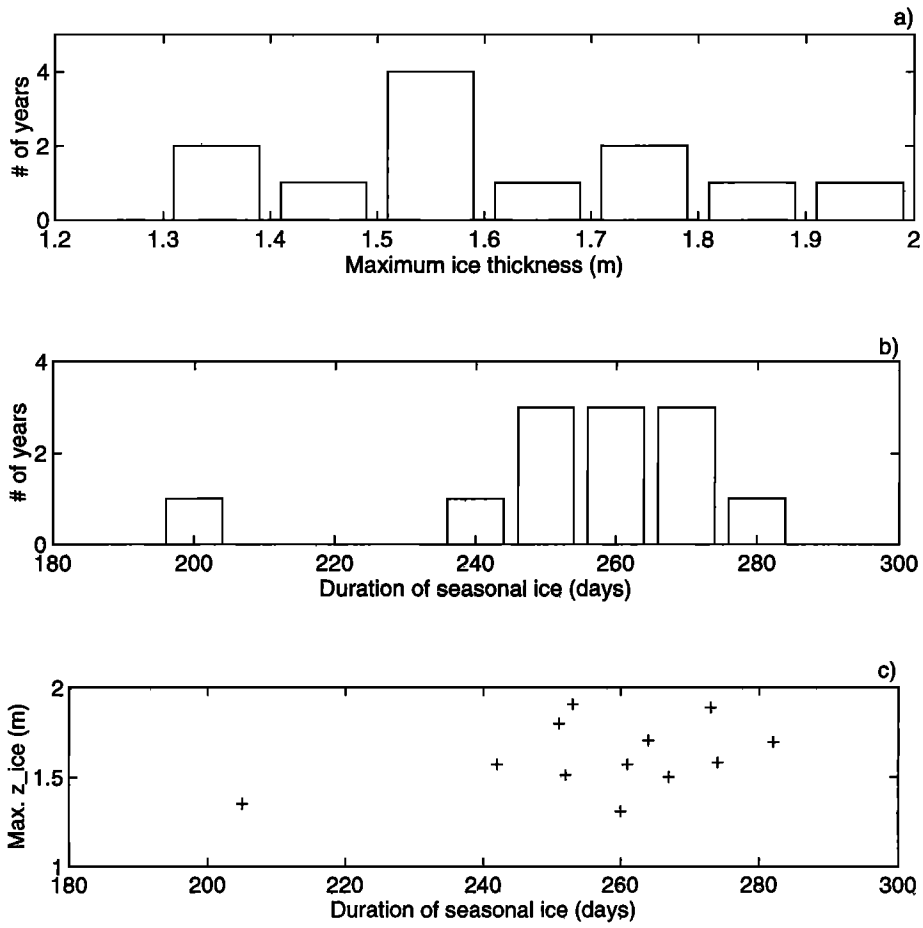

Figure 2. Statistics of seasonal fast ice formation at Mawson for 12 years from 1958 to 1986. (a) Maximum sea ice thickness, (b) Ice duration, and (c) maximum ice thickness as a function of duration.

values. For example, in spring 1965, typical measured values of the global radiation are only half of that which the model calculates. Although the results of the modified Zillman-type model do not agree well with the measured global radiation, we believe that the modeled results are more reliable. This is in agreement with *Shine* [1984], who found the accuracy of the incoming short wave radiation to be within ±5% if modeled with the modified Zillman parameterization (see section 2.1.1).

Also, the sensitivity of the thermodynamic ice model to the incoming short wave radiation is rather low. This is primarily because (1) the solar radiation during the ice growth season is low, and (2) the albedo of the ice surface is usually high, reducing the overall influence of the short wave radiation.

The maximum difference in the air temperature between the location of the weather station at Mawson and the locations of the ice measurement site is considered to be smaller than ±1 K. For 1969 this results in a change in maximum ice thickness of 4.4 cm (Figure 4a).

Figure 4c shows the response of the modeled ice thickness to variations in the oceanic heat flux of ±1 W/m^2 for 1969. While

Table 1. Maximum Ice Thickness Statistics for Mawson

Year	Start Day of Initial Ice Growth	Day of Midseason Breakout	Start Day of Second Ice Growth	Day of Maximum Ice Thickness	Maximum Ice Thickness, m	Maximum Snow Cover, m
1958	93	...	...	288	1.50	0.06
1965	94	130	135	274	1.35	0.07
1969	85	89	105	279	1.31	0.09
1976	88	...	...	279	1.57	0.05
1977	78	...	...	273	1.51	0.05
1979	82	88	101	288	1.71	0.02
1980	88	96	114	308	1.80	0.00
1981	104	...	...	294	1.57	0.05
1982	92	...	...	290	1.89	0.04
1984	85	89	91	297	1.58	0.04
1985	83	...	...	312	1.70	0.02
1986	82	85	102	295	1.91	0.03

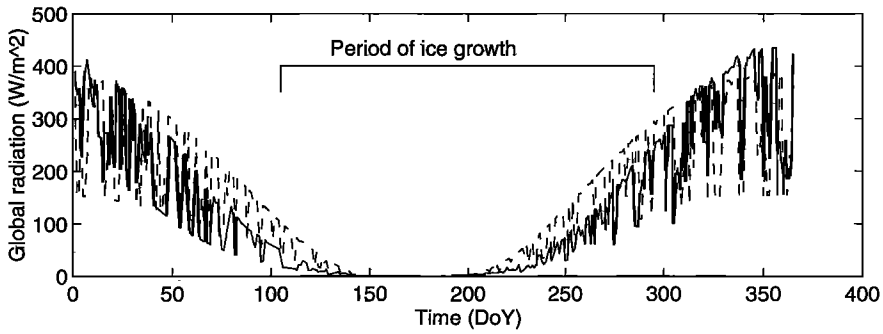

Figure 3. Comparison of the yearly distribution between measured and modeled global radiation for Mawson in 1965. Solid line shows the observed values, and dashed line shows the modeled radiation. Period of ice growth is indicated under bracket.

the ice is very thin ($z_{ice} < 0.5$ m), the growth rate and therefore the ice thickness are relatively insensitive to the oceanic heat flux which is only a small fraction of the total heat transport through the ice. However, for thicker ice the growth rate becomes sensitive to the magnitude of the oceanic heat flux. The maximum deviation between the standard and the "± 1 W/m^2" case is 5.8 cm. A 5% variation in the oceanic heat flux results at the end of the season in a 10.8% change in ice thickness.

The accuracy of the oceanic heat flux calculation depends on the growth rate of the ice. At constant accuracy of the ice thickness, considering the accuracy of the ice thickness measurements (± 2 cm), accuracy of the oceanic heat flux varies from ± 0.05 W/m^2 (growth rate < 0.18 cm/d) to ± 0.3 W/m^2 (growth rate > 3.1 cm/d). The accuracy of the oceanic heat flux increases as the ice growth rate slows. Table 2 summarizes the sensitivity of the maximum ice thickness and the modeled oceanic heat flux to short wave radiation, air temperature, and a number of other external forcing parameters.

From these tests we conclude that the growth of sea ice is highly sensitive to the incoming long wave radiation, the air temperature, and the oceanic heat flux. The strongest relative sensitivity is to the oceanic heat flux. This result encourages us to estimate the oceanic heat flux as the residual of the energy balance equation, when all other fluxes are derived from measured data. We conclude that by using this method the oceanic heat flux averaged over the whole ice season can be estimated to an accuracy of ± 1.0 W/m^2. Furthermore, because of the sensitivity of ice thickness to oceanic heat flux, we consider that the ice thickness cannot be satisfactorily modeled with a constant value of oceanic heat flux through the season. This is in agreement with *Washington et al.* [1976], who found for the Arctic that ice thickness is quite sensitive to seasonal changes in the oceanic heat flux.

5. Results

For the 12 years for which detailed ice thickness measurements around Mawson are available the oceanic heat flux was determined as the residual of the energy balance combined with ice thickness data. This provides an estimate of the annual variability of the oceanic heat flux.

We choose 1969 as reference year because the frequency of the ice thickness measurements was the highest. The calculated oceanic heat flux and the measured ice thickness are shown in Figure 5. It is obvious from these results that the oceanic heat flux can be subdivided into three different time regimes. Until the end of August (day of year (DOY) 240) the oceanic heat flux averages 7 W/m^2. From late August until the end of September (DOY 270) the heat flux averages 12 W/m^2 and increases after this to an average of 18 W/m^2. *Allison* [1981] found two peaks in the seasonal cycle of the oceanic heat flux. The first peak, explained by strong thermohaline convection, proved to be highly overestimated, since the snow cover was not properly accounted for. The second peak matches well with our results: For 1969, *Allison* [1981] obtained an oceanic heat flux of 18.8 W/m^2 in late September, which

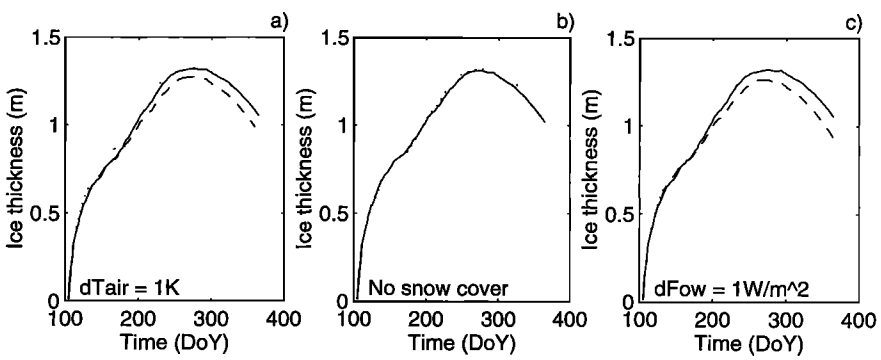

Figure 4. Sensitivity of the thermodynamic model to external forcing parameters for 1969. Standard case is the solid line. (a) Change in air temperature of 1 K increase (dashed line) and 1 K decrease (dotted line), (b) no snow cover at all (dotted line), and (c) influence of the oceanic heat flux of a 1 W/m^2 increase (dashed line) and a 1 W/m^2 decrease (dotted line).

Table 2. Seasonal Sensitivity of the Maximum Ice Thickness and the Derived Oceanic Heat Flux to External Model Forcing

Parameter	Typical Annual Average	Range of Variations From the Actual Value	$\lvert\delta h_{ice_{max}}\rvert$, cm	$\lvert\delta F_{ocean}\rvert$, W/m^2
SW_{in}	138 W/m^2	±5%	1.3	0.20
I_0	17%	±100%	0.6	0.09
LW_{in}	198 W/m^2	±5%	5.6	0.87
T_{air}	261 K	±1 K	4.4	0.68
q_{air}	0.83	±5%	1.7	0.26
u	12.5 m/s	±50%	0.8	0.13
Snowcover	0.05 m	−100%	1.6	0.25
F_{ocean}	11.6 W/m^2	±1 W/m^2	5.8	...

compares with our calculated value of 18 W/m^2 at the end of September just before the maximum ice thickness was reached.

Because the ice thickness data from years other than 1969 are less frequent, we present an averaged oceanic heat flux for different time regimes rather than at each time step. Prior to 1979 the derived values of oceanic heat flux show a similar seasonal variation each year with significant increases occurring at the end of August (DOY 240) and the end of September (DOY 270) (Table 3). After 1979 the derived oceanic heat flux varied in a more complicated manner, with up to four changes in magnitude required to match the measured ice growth. For 1979 and later the heat flux early in the year is significantly reduced, remaining below 4 W/m^2 at least until DOY 240 and in many years until DOY 270. With exception of 1979, in all years the oceanic heat flux reaches at least 15 W/m^2 by the end of the year, although after 1979 this increase occurs later by as much as 1 month. In order to compare between the observed years we calculate the oceanic heat flux averaged over the three fixed periods (Table 4). A significant decrease in the heat flux regime at the end of the 1970s is evident.

In order to compare the average annual oceanic heat fluxes we calculate the total energy as the oceanic heat flux multiplied by the period in days, over which this flux is applied (Table 5), and the weighted average oceanic heat flux is derived from this. The comparison over all years for which data are available indicates a long-term trend: The average oceanic heat flux decreases from 1958 to the 1980s (Figure 6). This coincides with higher ice thickness maxima for the years in the 1980s than in the earlier years. Figure 7 shows the relationship between the maximum ice thickness and the calculated oceanic heat flux.

As a check on our calculations, we also model the oceanic heat flux using data on ice thickness and forcing parameters from Davis for the years from 1980 to 1985. Because the bathymetry around Davis is rather shallow and the coastal waters are not subject to exchange with the deeper waters, we do not expect high values for the oceanic heat flux at Davis. The results confirm this: Figure 6 shows that the oceanic heat flux is small compared to the values that were obtained for Mawson.

6. Discussion

6.1. Seasonal Variability

The high sensitivity of the thermodynamic model to the oceanic heat flux allows an estimation of the seasonal changes in the oceanic heat flux. The general seasonal feature for all years reveals relatively low values of the oceanic heat flux in the early growth period, typically <6 W/m^2. In late winter (DOY 240–270) the heat flux increases, often doubling over the early values. The seasonal variability falls into two distinct regimes. In the earlier years (1958–1977) the oceanic heat flux increases from 6 W/m^2 to ~10–12 W/m^2 at the end of August (DOY 240) with a further increase a month later to values of 15–18 W/m^2. The seasonal variation of the oceanic heat flux is

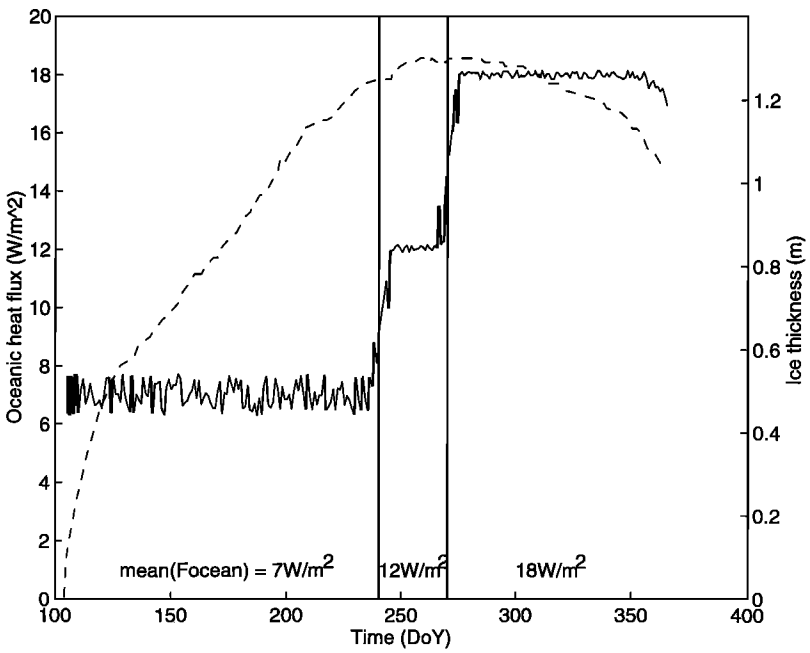

Figure 5. Seasonal development of the sea ice thickness for 1969. Dashed line indicates the observed ice thickness, while solid line represents the modeled oceanic heat flux. Averaged oceanic heat flux (in watts per square meter) averaged for the three characteristic regimes is indicated by vertical lines.

Table 3. Variations of the Oceanic Heat Flux Required for Best Fit of the Multilayer Model

Year	F_{ocean_1}, W/m^2	DOY_1	F_{ocean_2}, W/m^2	DOY_2	F_{ocean_3}, W/m^2	DOY_3	F_{ocean_4}, W/m^2	DOY_4	F_{ocean_5}, W/m^2
1958			6	240	10	270	18		
1965			6	240	12	270	18		
1969			7	240	12	270	18		
1976			5	240	12	270	17		
1977			6	240	10	270	17		
1979	2	200	0	240	4	270	13		
1980	0	190	5	240	8	270	2	310	18
1981	0	190	6	240	10	270	0	280	18
1982			2	240	0	280	10	300	15
1984	0	190	1	240	8	270	18		
1985	0	160	1	190	4	240	1	290	18
1986	4	150	1	190	0	240	5	295	15

DOY_i is the day of the year for which a change in the oceanic heat flux occurred in order to match the measured ice growth.

steady and pronounced in the early years, while in the more recent years the heat flux tends to fluctuate without following a specific pattern, although there is still a tendency to increase from initial values <6 W/m^2 to 18 W/m^2.

There are two potential sources of the oceanic heat flux. First, absorption of solar radiation in the surface layer in times of ice-free ocean could provide heat which is then transported horizontally. The other source for oceanic heat into the upper layer of the polar ocean is by the upwelling of warmer deep water. *Gordon* [1981] estimates that both mechanisms on average over the year contribute equally to the oceanic heat in the upper ocean of Southern Ocean regions. The input of solar radiation is dominant during summer months, while the upwelling of deep water is stronger in wintertime. Since the penetration of deep water from the deep ocean onto the shelf takes place in certain locations only, the winter supply of oceanic heat from the deep ocean onto the shelf is confined to few source locations, for example, Prydz Bay.

Within Prydz Bay itself there is a cyclonic gyre, which extends north over the shelf break. Most of the inflow to the region is from the north and the west [*Nunes Vaz and Lennon*, 1996; *Smith et al.*, 1984]. The outflow of the gyre hugs the western rim of the bay, and as it exits the bay, it travels westward along the shelf. Some of this coastal current is diverted north to rejoin the Antarctic Circumpolar Current (ACC), with the rest continuing westward along the shelf (Figure 1). The relative magnitude of these two branches of the coastal current is uncertain. The waters in our study region are at least partially formed from this shelf current which has traveled through the Prydz Bay region. This is confirmed from hydrographic surveys [e.g., *Smith et al.*, 1984], studies of iceberg and drifting buoy tracks [*Allison*, 1989], and biological studies [*Hosie*, 1991]. A mean transport speed of 0.2 m/s was derived for the coastal current between Prydz Bay and Mawson from both biological studies and iceberg drift. Most of oceanographic data in this region were collected during summer months. However, *Allison et al.* [1985] report on conductivity-temperature-depth (CTD) data collected during the winter on the continental shelf near Mawson. The data were collected over 2 different years (1980 and 1982), but together they comprise a composite picture of an annual cycle. From early July (DOY 190) to late November (DOY 330) the water is near isothermal at the freezing point (−1.8°C). There is no excess heat in the water to supply to the ice, and during this period the oceanic heat flux is <2 W/m^2. As spring approaches, the water temperature increases markedly as a result of advection, and the increase in heat content of the water follows the same general pattern as the calculated oceanic heat flux (Figure 8). Hence we believe that the changes in oceanic heat flux are indicative of seasonal changes in the water masses advected along the shelf. A similar pattern occurs in the other years also, with an increase in oceanic heat flux each spring from DOY 270 (end of September) to DOY 303 (end of October). However, the magnitude of this increase is more abrupt after 1979 than in the earlier years. With the exception of 1981 we see for years later than 1979 an increase in the oceanic heat flux from

Table 4. Interannual Change of the Oceanic Heat Fluxes Weighted Over the Three Defined Periods

Year	Period 1 DOY < 240	Period 2 240 < DOY < 270	Period 3 DOY > 270
1958	6.0	10.0	18.0
1965	6.0	12.0	18.0
1969	7.0	12.0	18.0
1976	5.0	12.0	17.0
1977	6.0	10.0	17.0
1979	1.4	4.0	13.0
1980	2.0	8.0	11.3
1981	2.2	10.0	16.1
1982	2.0	0.0	12.4
1984	0.3	8.0	18.0
1985	1.5	1.0	14.4
1986	1.7	5.0	12.4

Interannual change is in watts per square meter. DOY, day of year.

Table 5. Total Energy and Weighted Average Oceanic Heat Flux

Year	Duration of Seasonal Ice Cover, days	Energy, 10^6 J/m^2	Weighted Oceanic Heat Flux, W/m^2
1958	267	243.7	10.6
1965	205	196.0	11.1
1969	260	262.1	11.7
1976	242	186.4	8.9
1977	252	199.5	9.2
1979	264	134.2	5.9
1980	251	134.8	6.2
1981	261	158.6	8.2
1982	273	127.1	5.4
1984	274	174.4	7.4
1985	282	142.6	5.8
1986	253	133.1	6.0

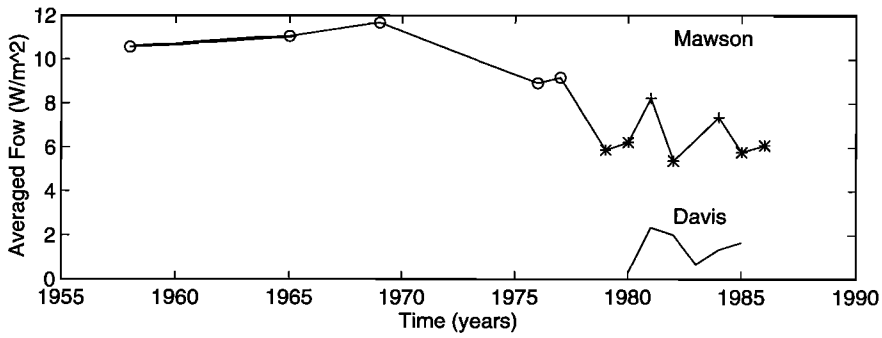

Figure 6. Time series of the averaged oceanic heat flux. Circles indicate the years for which no passive microwave imagery is available. For later years, SMMR data provide information on the occurrence of polynyas. There is a correlation between low oceanic heat fluxes around Mawson and the existence of a polynya (indicated by stars) in the Prydz Bay region. Years with no polynya (indicated by crosses) coincide with higher values in the oceanic heat flux.

<8 W/m^2 to 15–18 W/m^2. In 1981 and the years prior to 1980 this increase is more gradual from around ~10–12 W/m^2 in September to 13–18 W/m^2. This may indicate that prior to 1980 the water advected along the shelf during September was warmer than what it has been since that time.

Offshore from Mawson at 630-m water depth, current meter data from 1985 [*Nunes Vaz and Lennon*, 1996] give evidence of modified warm Circumpolar Deep Water escaping the Prydz Bay gyre and propagating toward the northwest at the beginning of spring. Currents were measured for an entire year at depths of 325, 516, and 620 m; an additional current meter at 178 m operated for a short time only. The current direction at all depths was consistently to the west northwest, with speeds of up to 1.2 m/s, an order of magnitude higher than in Prydz Bay itself. The exception to this was during July and August when the currents decreased significantly. It is interesting to note that the currents were not correlated with the winds at Mawson station. The northwestward flow of the coastal current along the shelf offshore from Mawson is in agreement with the distribution of the geopotential anomaly as derived from hydrographic data [*Nunes Vaz and Lennon*, 1996]. Additional current meter data from within Prydz Bay consistently shows reduced flow from July to early August at all depths. Because of these reduced flow rates we estimate that the water resides 3–5 times longer within Prydz Bay than at other times. Consequently, this water is exposed to any cooling processes taking place in Prydz Bay for a longer time. From a combination of the current speed along the shelf and the reduced flow within Prydz Bay we conclude that water, which experienced this extended cooling in Prydz Bay during July and early August, would have passed Mawson by the end of September and warmer water would follow. Our model results for 1985 give an oceanic heat flux of 4 W/m^2 or less until DOY 290 (October 17), increasing to 18 W/m^2 after that date. From the limited data available we hypothesize that the seasonal variability of the oceanic heat flux may be caused by changes in the properties of the water bodies. These changes may result from slackening in the current speed. The circulation pattern appears to be unchanged.

Since we consider the Prydz Bay region as the source of these changes in water bodies reaching the shelf offshore Mawson, we investigate the processes which induce cooling on the water masses in Prydz Bay. Active sea ice formation enhances the thermohaline convection in the winter months, cooling the intrusions of Circumpolar Deep Water (CDW), especially during the beginning of the ice growth season. Strong cooling will continue in regions of open water, particularly in polynyas. Generally, in November as air temperatures warm the sea, ice concentration in the Prydz Bay begins to decrease below 60–80% [*Gloersen et al.*, 1992]. The reduction in ice formation and the associated decrease in the vertical thermohaline convection may allow CDW to reach further onto the shelf or, alternatively, to travel within the general circulation pattern through the Prydz Bay region without significant cooling, allowing warmer water to reach the shelf near Mawson. There is insufficient year-round hydrographic data in the pack ice zone

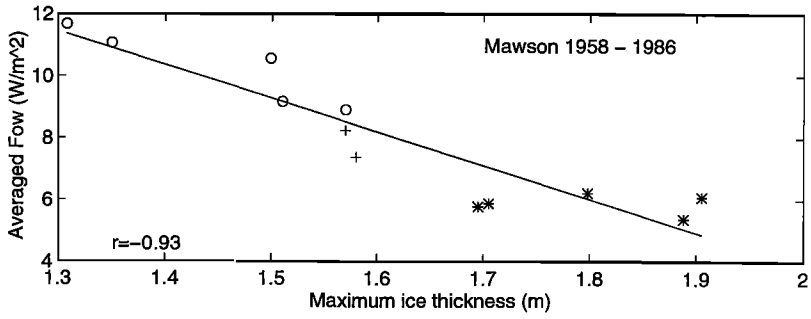

Figure 7. Relationship between the yearly maximum ice thickness and the weighted oceanic heat flux. Symbols are the same as in Figure 6. A linear regression through the data yields to a regression coefficient of 0.93.

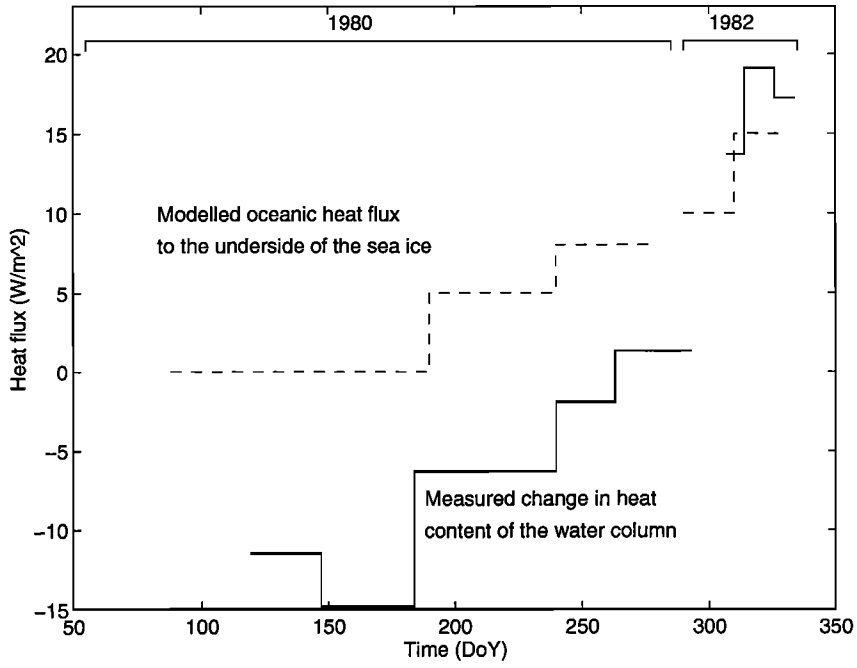

Figure 8. Change in heat content of the water column calculated from CTD profiles in 1980 and 1982. Change of heat content (solid line) of the water column is in phase and of the same order of magnitude as the gain of oceanic heat (dashed line) by the sea ice estimated from our model.

to confirm this, but the good correlation between the onset of the sea ice melt as observed at Mawson and the increase in the oceanic heat content as derived from available CTD measurements is encouraging. Increased open water areas starting in the middle of spring will also allow increased absorption of solar radiation. Consequently, the warmer waters seen around Mawson and the associated increase in the oceanic heat flux are likely the result of the decrease in ice formation and pack ice concentration in the Prydz Bay region.

6.2. Interannual Variability

The average oceanic heat flux has decreased since 1958 (Figure 6). Before 1979 the annual average oceanic heat flux ranged ~11 W/m^2, with the total flux decreasing by 2 W/m^2 in the mid-1970s. From 1979 onward we find consistently low values for the total oceanic heat flux, averaging ~6.4 W/m^2.

Passive microwave satellite data show that the Prydz Bay region is the location of the greatest seasonal and interannual variability in the East Antarctic sea ice concentration. Analysis of passive microwave data of the Nimbus 7 Scanning multichannel microwave radiometer (SMMR) for the years 1979–1986 reveals that the longest sea ice season for the area of the Indian sector off East Antarctica was in 1981 [*Parkinson*, 1994]; the shortest season within this period was in 1986. The same microwave data shows that the maximum extent in the sea ice corresponds to the longest season (1981), while the minimum extent coincides with the shortest season (1986). The maximum ice thickness at Mawson was 1.57 m during 1981 and 1.91 m for the 1986 season, the thickest fast ice measured after 1979. From our model we found that the averaged oceanic heat flux was 8.2 W/m^2 in 1981 and 6.0 W/m^2 in 1986. The years with more sea ice in Prydz Bay correspond to a thinner fast ice cover and higher oceanic heat flux near Mawson. Similarly, years with a shorter ice season in Prydz Bay correspond to a thicker fast ice cover and a smaller oceanic heat flux near Mawson.

Polynyas occur frequently in the southwest side of Prydz Bay [*Gloersen et al.*, 1992]. These are most likely generated and maintained by strong and persistent offshore winds in Mackenzie Bay. To quantify the persistence of the polynya in the southwest corner of Prydz Bay, we analyze passive microwave data (C. L. Parkinson and J. Saleh, NASA Goddard Space Flight Center, unpublished data, 1979–1986). Additional data are available from the National Snow and Ice Data Center (NSIDC) World Wide Web site at http://www-nsidc.colorado.edu/NASA/GUIDE/docs. We calculate the average number of days with sea ice of 50% or greater concentration present both in this part of Prydz Bay and in the rest of Prydz Bay. The difference between these values is the length of time the polynya was present. Figure 9 shows the yearly averaged oceanic heat flux against this difference in ice duration. Similar scatterplots are seen for 30 and 15% concentration. Only in 1981 and 1984 was the ice duration (and concentration) in the southwest corner similar to the rest of the bay, indicating no polynya formed during those years. In other years the ice cover was far less persistent in the southwest corner. For years other than 1981 and 1984 a patch of low ice concentration in winter persisted for a total of 47–61 days.

Openings or polynyas in the sea ice cover of southwest Prydz Bay coincide with very thick fast ice at Mawson, while in years without a winter opening in southwest Prydz Bay this area had thinner ice (Figure 6). The oceanic heat flux in years without an open water period in Prydz Bay is 35% higher than in years with an opening. A possible coupling mechanism between the two phenomena is the very strong loss of heat from the ocean into the atmosphere over the open water area. Also, salt rejection as new ice forms causes deep thermohaline convection, which enhances the rise of warm, deep water into the mixed layer [*Oberhuber et al.*, 1993]. This deepens the mixed layer, reducing the amount of warm, deep water which arrives

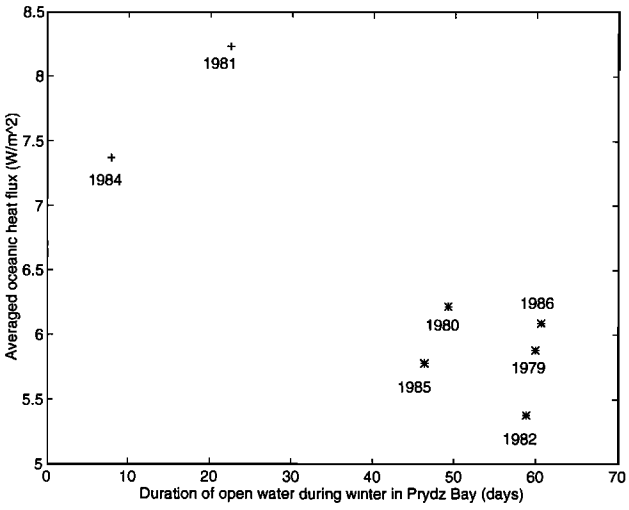

Figure 9. Yearly average oceanic heat flux against the regional difference in the duration of sea ice coverage of 50% or greater concentration. Regional difference is the difference between ice duration for all of Prydz Bay except the southwest corner and the ice duration in the southwest corner (Mackenzie Bay region) only. Symbols are the same as in Figure 6.

"downstream" at Mawson. This is in agreement with the model estimations of the oceanic heat flux; increased polynya duration results in decreased oceanic heat flux (Table 5).

Before 1979 there are no SMMR data to determine periods of ice coverage and concentration of the sea. We hypothesize that the long-term decrease in oceanic heat flux at Mawson is linked to an increase in the occurrence of areas of open water within the winter pack in Prydz Bay. Comparing the interannual variations in oceanic heat flux between the 1950s and the 1980s, we conclude that there were fewer open water events in (southwest) Prydz Bay region in the late 1950s and 1960s. This may imply that there has been a change in the atmospheric circulation, since the open water area in Prydz Bay is mainly generated by winds, or that there has been a change in the water properties of the region.

7. Conclusions

We find strong evidence for a seasonal variation in the oceanic heat flux in the near-offshore area around Mawson. The oceanic heat flux must increase threefold or even more in later years at the end of winter to explain the observed decrease in ice thickness. The seasonal cycle of sea ice formation and decay in Prydz Bay may be a plausible cause of the seasonal variation in the oceanic heat flux downstream of the Prydz Bay gyre, near Mawson. The thermohaline convection caused by the ice formation and associated brine rejection cools the deep water as it flows through the gyre. This is analogous to the larger Weddell gyre where CDW is cooled as it traverses the gyre [*Fahrbach et al.*, 1994]. Consequently, early in winter, when the sea ice formation rate in Prydz Bay is high, thermohaline convection may cool the deeper, warmer waters so that the waters near Mawson are near-isothermal at −1.8°C, and there is little oceanic heat flux. The approach of spring, with little or no sea ice growth in Prydz Bay, corresponds to an increase in the water temperatures around Mawson and an increase in the oceanic heat flux. Therefore the oceanic heat flux under the fast ice is influenced by deep convection and mixing which occurs as the water travels through the gyre. Increased absorption of solar radiation in spring also contributes to this warming.

One of the difficulties in the analysis of long-term changes in the sea ice conditions in the Antarctic is the lack of a longtime history of data on sea ice conditions. Not until the early 1970s were routine data on sea ice extent available. Although it is not as extensive or comprehensive as data obtained using satellite sensors, the data set presented here extends more than a decade prior and allows us to identify large-scale changes in the oceanic regime in the East Antarctic region.

The maximum thickness of the fast ice near Mawson has increased from ~1.3 m in the late 1950s to ~1.8 m in the mid-1980s. This corresponds to almost a 50% decrease in the average oceanic heat flux from ~11 to 6 W/m^2 over the 3 decades of our study. Although these data are from a single location, they are indicative of broader changes which are occurring in the Prydz Bay region. We suggest that this decrease in oceanic heat flux is related to decadal changes in sea ice conditions in Prydz Bay. In recent years, when we can compare the oceanic heat flux with the sea ice conditions using passive satellite data, there is a good correspondence between a lower oceanic heat flux at Mawson and less sea ice in Prydz Bay. This change could also be the result of changing properties in the oceanic circulation, for example, less (or cooler) CDW entrained in the Prydz Bay gyre. However, the agreement of the seasonal variations in both Prydz Bay (extensive ice formation or polynya events) and offshore Mawson (low oceanic heat flux at the beginning of the ice growth season or especially low oceanic heat flux after the initial ice growth) point to a causal connection between the decrease in the sea ice at Prydz Bay and the changes in the landfast ice at Mawson.

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